Highlights

When does adaptive correction of day-ahead forecasts pay? Evidence from India's power system

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- One fixed gate cut State drawal error energy by 13,374 GWh/yr (post-hoc analysis)
- Finer gates added error in 9 of 36 Indian State cases and removed it in 3
- Validation-only gate choice was not confirmed on Indian data (5 of 20 tests)
- Bias correction refitted daily beat learned gates for 4 of 8 all-India forecasts
- Correction value changed sign between stylised and DSM-style settlement rules

When does adaptive correction of day-ahead forecasts pay? Evidence from India's power system

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ARTICLE INFO

Keywords:
Load forecasting
Forecast correction
Indian power system
Operating reserve sizing
Imbalance cost
Estimation error
Pre-registered evaluation

ABSTRACT

Day-ahead demand, drawal and generation forecasts are corrected with models of recent errors, and the weight (gate) on the correction is increasingly specific to a region, hour, observation or time. As forecast errors must be balanced or covered by reserve, we ask when such adaptivity pays in energy terms in India. A finer gate helps only if the variation it exploits is large, persists into operation and outweighs its estimation cost; two validation-only rules estimate this criterion, tested under two pre-registrations on government data. On 34 State control areas (April 2025 to August 2026), one fixed gate removed 13,374 GWh per year (15 %) of next-day drawal forecast-error energy (post-hoc analysis), whereas finer gates left significantly more error energy in 9 of 36 cases and less in 3. On all-India hourly series, the validation-only rules, confirmed earlier on New York, European and client data, were not confirmed (5 of 20 tests supported), and no design helped the N-HiTS or Chronos-Bolt forecasts. In exploratory benchmarks, a simple bias correction refitted daily on each forecast's own errors removed 34.7 % of the squared error of a LightGBM forecast, against at most 12.7 % for any gated design, and gating on top of it added little. Correction lowered forecast-error energy but raised a stylised two-settlement imbalance cost, whereas a deviation-settlement-style charge valued it positively. For Indian control areas, a simple correction refitted on recent errors is the safer default; finer or learned gates need evidence that the structure they exploit persists.

1. Introduction

Day-ahead forecasts of demand, drawal and generation drive unit commitment, the energy that is bought before the day and the operating reserve a system operator holds against forecast error, a need that grows with the share of renewable generation [1]. In India, every State control area schedules its drawal from the grid a day ahead and pays for deviating from that schedule under the deviation settlement [2]; in markets such as New York, a load-serving entity buys in the day-ahead market and settles the difference in real time. Forecasts are rarely used as issued. Operators and forecast vendors correct them with models of recent errors, and machine-learning pipelines add correction layers on top of statistical, deep-learning and foundation forecasters. Every such layer has a *gate*, a weight that decides how much of the learned correction to apply. The prevailing design instinct is to make the gate as adaptive as possible: one weight per region, per regime, even per observation, re-estimated as new data arrive.

In India each State is a control area. Its State Load Despatch Centre (SLDC) must estimate demand a day ahead in 15-minute time blocks, send the daily estimate to its Regional Load Despatch Centre by 10:00 hours of the previous day, and compute and publish its day-ahead and intra-day forecast errors; the regulation leaves the method to each SLDC [3, Regulation 31]. A State's deviation, its actual minus scheduled drawal in each time block, has been settled since 16 September 2024 under the Deviation Settlement Mechanism Regulations, 2024 [2, 4], at a normal rate equal to the highest of the day-ahead and real-time market prices and an average of these with the ancillary-service charge, scaled with grid frequency and subject to volume limits. Secondary reserve requirements may be set from the 99th percentile of the area control error or from the largest unit plus the previous year's load, wind and solar forecast errors [3, Regulation 30(11)]. The system is large and changing: all-India peak demand met reached 256.1 GW on 25 April 2026, when solar plants supplied 56.2 GW [5].

This paper asks when the adaptive instinct is right, and what the answer is worth in operation. A finer gate can only exploit variation that exists in the correctable part of the error, persists from the period in which the gate is estimated to the period in which it is used, and can be estimated from the data available; when any of these fails, a more adaptive

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correction is less accurate, and the loss propagates to the energy that must be balanced and the reserve that must be held. We therefore value each design in energy, reserve and settlement terms, not by accuracy alone.

We make four contributions.

1. **A testable criterion for adaptive corrections.** Known results on forecast combination and estimation error are combined into one criterion: refining a gate pays only if the variation it exploits is large, persists into deployment and outweighs its estimation cost (Section 3). Two rules estimate the criterion from validation data only, one for regional and regime refinements and one for adaptation over time, and the Indian studies test whether they work.

2. **Evidence on Indian State control areas.** On a State-day panel of 34 Indian control areas built only from official reports, for energy met, drawal, conventional and renewable generation and the day-ahead price, we test the validation-only rule under that study's pre-registration and, in an additional analysis, value every gate design in forecast-error energy, in GWh per year, and for the price in Rs/MWh (Section 5).

3. **A pre-registered test on the all-India hourly series.** From the 15-minute SCADA values in Grid-India's daily reports we build hourly series of all-India demand met, net demand, wind and solar generation, correct eight day-ahead forecasts, from weekly persistence to a foundation model, and test the selection rules, the uncertainty models and the energy value of correction in 20 tests frozen before any forecast for the test months was made.

4. **Benchmarks and robustness checks on Indian data.** Under a protocol committed before they were run, we compare the gated corrections with the bias corrections operators use, refresh the base forecast, move the validation window up to each month of use, recalibrate the reserve quantiles and vary the settlement rule.

5. **A comparison with four other systems.** The same method, pre-registered earlier on New York (NYISO) zonal load, 46 European transmission-system operator (TSO) forecasts, 60 electricity clients and a semi-synthetic grid, shows which findings carry over to India and which do not.

On both Indian datasets, finer gates were not reliably better than one coarse gate, the validation-only rules that worked on the comparison data did not tell in advance when a finer or faster-adapting gate would pay, and a simple bias correction refitted on recent errors did better than any learned gated correction. Lower forecast error did not by itself lower the stylised imbalance cost, while a charge in the style of India's deviation settlement valued it positively; the evidence speaks to State and regional load despatch centres and to the regulator (Section 6).

2. Literature review

Forecasting for Indian control areas. Day-ahead demand forecasting at Indian State load despatch centres has been studied with neural networks on SLDC data [6], long-term demand projections for India now account for electric-vehicle charging and space cooling [7], and the economics of solar forecasting under India's deviation settlement have been modelled [8]. The rapid growth of renewable generation raises the stakes of these forecasts [1], and how forecast errors in an emerging-economy system translate into dispatch cost and emissions is an open question [9]. We did not find a study that tests, on official Indian data and before the test data are seen, how finely a correction of such forecasts should adapt.

Correcting operational forecasts. Post-processing a forecast with a learned model of its errors is standard practice in energy forecasting [10, 11], from system-level load forecasts to substations [12], and recent work adds safeguards to post-hoc corrections of deep forecasters [13]. Many forecasting systems in this journal add an explicit error-correction stage to a primary model, for hourly solar irradiance [14], for wind power with a linear error correction [15] and, with model selection inside an ensemble, for interval-valued wind speed [16]; ensemble model output statistics calibrate wind forecasts [17], and for photovoltaic forecasts made from ensemble weather predictions bias correction was the most important post-processing step [18]. Operators' published forecasts are corrected in the same way: Möbius et al. [19] remove the bias of the German transmission-system operator's day-ahead load forecast with an hour-of-week mean and an autoregressive remainder re-estimated on a rolling window, and Xie et al. [20] study bias calibration of the PJM load forecast. Adaptive probabilistic load forecasting that tracks distribution shift is a related line [21]. Strong deep-learning and foundation forecasters [22–27] and gradient boosting [28] raise the bar that any correction must clear.

Table 1
Lines of work on correcting energy forecasts, how each sets the adaptivity of the correction, and what this paper adds.

Line of work	Examples	Adaptivity of the correction	Added here
Error-correction stages in hybrid forecasters	[14–16]	fixed by design	a test, before deployment, of when a finer correction pays
Bias calibration of operator forecasts	[18–20]	hour-of-week or rolling window, judged on test accuracy	the same calibrations as benchmarks on Indian series
Adaptive and online forecasting	[21, 54]	continuous	when adaptation over time pays, with India's two-day publication lag
Forecast combination	[49–51]	estimation error and instability explain why simple weights win	an estimable crossing condition and two validation-only rules
Quantiles for reserve	[30, 34, 35]	conditioning cells, adaptive calibration	the tail cost of fine conditioning; calibration in the season of use
Value-oriented forecasting	[41, 42, 47]	training for the decision	accuracy gains and settlement value compared under Indian prices
Forecasting for Indian control areas	[6–8]	–	pre-registered tests on official Indian government data

Reserve sizing from forecast errors. Dynamic and probabilistic dimensioning of operating reserves from forecast errors [29–31] and reserve scheduling under uncertainty [32] need reliable high quantiles of the error distribution, for example from quantile-regression networks [33]. Split conformal quantiles [34] are calibrated when calibration and use are exchangeable; under distribution shift, adaptive and weighted conformal methods restore calibration [35–38], also for electricity prices [39], and proper scoring rules evaluate the result [40]. How finely such quantiles should be conditioned is the reserve-side instance of the question studied here.

Forecast value versus forecast accuracy. Accuracy measures often fail to reflect the economic value of forecasts in operation [41, 42]; the value of photovoltaic forecasts depends on the trading and scheduling rules they feed [43, 44], imbalances can themselves be forecast [45], and value-oriented forecasting trains the forecast for the decision it serves [46–48].

Why simple weights often win. A correction gate combines the original and the corrected forecast. The forecast-combination literature, from Bates and Granger [49] to the review of Wang et al. [50], explains why estimated weights often lose to simple ones, through estimation error [51] and instability over time [50]. The criterion used here makes both explanations measurable for zone, hour, observation and time adaptivity, building on covariance penalties [52] and on the tracking trade-off of adaptive filters [53]. Table 1 places the paper among these lines of work.

3. Methodology

3.1. Gated correction and the adaptivity ladder

Let Y be the target series (load, drawal, generation or price), B a baseline forecast and $R = Y - B$ its error. A corrector $\hat{r}(x)$, gradient-boosted trees trained on a separate training period, predicts R using only information available when the day-ahead forecast is issued (observations up to the end of day $D-2$ for target day D). On the all-India hourly series its features are the values of Y 48, 72 and 168 hours earlier, the 168-hour rolling mean and standard deviation of Y ending 48 hours earlier, hour, weekday, day of year and a series code; it does not use B or its past errors. On NYISO the features also include the ISO forecast and its lagged errors. $\hat{r}$ is rescaled by one least-squares factor on the gate-fitting set before the gates are fitted. The corrected forecast is $B + g\,\hat{r}(x)$, where the gate g is estimated by least squares on a gate-fitting period (the validation period) and clipped to $[0, 1.5]$; a cell with fewer than 30 observations takes its parent's gate. The propositions below describe unclipped least-squares gates; clipping only bounds extreme estimates. The gate may depend on an *adaptivity index*. We compare a ladder of designs along three axes:

- **unit**: no correction; one global gate; one gate per zone (area, client, series);
- **state**: one gate per zone and 4-hour block or per zone and hour; a per-instance gate that is a smooth function of the features (three bagged gradient-boosted models fitted by weighted least squares), and its shrunk version;

- **time**: the frozen per-zone gate; the same gate updated daily with realised errors (expanding window); rolling windows of 30, 90 and 180 days; exponential forgetting with daily factors 0.98 and 0.995; and Fixed Share aggregation of baseline and correction [54]. Updates use errors realised two days before the target day.

All hyperparameters were fixed before any evaluation and are identical across systems.

Figure 1 summarises the procedure: the corrector, the ladder of gate designs and the two rules that choose a design from validation data.

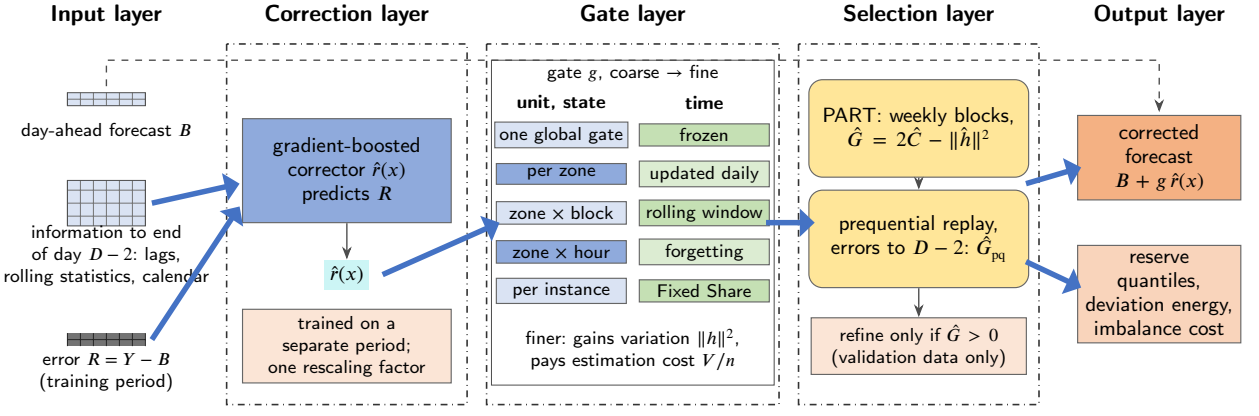


Figure 1: The method in layers. A corrector trained on a separate period predicts the error of the day-ahead forecast from information available at issue time; a gate scales the correction, and gate designs become finer along the unit, state and time axes; two rules choose a design from validation data only, by estimating the crossing condition of Proposition 4. The dashed line carries the original forecast B.

3.2. The price of adaptivity

A gate design assigns one gate to each cell c of a partition $\mathcal{P}$ of the observations (zones, or zones and hours); the per-instance gate is the limit of fine partitions, and adaptation over time is treated in Proposition 5. For a distribution Q of (x, R), either the gate-fitting distribution V or the deployment distribution T, let $W_Q(c) = \Pr_Q(c)\,\mathbb{E}_Q[\hat{r}^2 \mid c]$ be the correction energy of cell c, and for functions constant on cells write $\langle f, h\rangle_Q = \sum_c W_Q(c) f(c) h(c) = \mathbb{E}_Q[\hat{r}^2 f h]$ and $\|f\|_Q^2 = \langle f, f\rangle_Q$. Throughout, (A0) $\mathbb{E}_Q[R^2] < \infty$ and $0 < \mathbb{E}_Q[\hat{r}^2 \mid c] < \infty$ in every cell. The oracle gate $g^\star_{\mathcal{P},Q}(c) = \mathbb{E}_Q[R\hat{r} \mid c]/\mathbb{E}_Q[\hat{r}^2 \mid c]$ then minimises the risk $L_Q(g) = \mathbb{E}_Q[(R - g\hat{r})^2]$ over gates constant on the cells of $\mathcal{P}$, and a refinement $\mathcal{P}'$ of $\mathcal{P}$ can exploit the heterogeneity $h_Q = g^\star_{\mathcal{P}',Q} - g^\star_{\mathcal{P},Q}$. None of the results below is new in isolation; the contribution is to combine them into a criterion that can be estimated from validation data, and to test it. Each result is marked as an identity, as exact under stated assumptions, or as asymptotic; proofs are in Section A.

Proposition 1 (Risk difference; identity)**.** *For gates g_a, g_b with $d = g_b - g_a$ and $e_a = R - g_a\hat{r}$, and any distribution with finite second moments, $L(g_a) - L(g_b) = 2\mathbb{E}[d\hat{r}e_a] - \mathbb{E}[d^2\hat{r}^2]$.*

Proposition 2 (Heterogeneity gain; exact under (A0))**.** *Refining $\mathcal{P}$ to $\mathcal{P}'$ lowers the oracle risk by $L(g^\star_{\mathcal{P}}) - L(g^\star_{\mathcal{P}'}) = \|h\|^2 \geq 0$.*

Proposition 3 (Estimation cost)**.** *Let the cell gates be least-squares estimates $\hat{g}_c = \sum_{i\in c} R_i\hat{r}_i / \sum_{i\in c}\hat{r}_i^2$ from n fitting observations, scored on independent data from the same distribution. (a) Exact for a fixed design: if $R_i = g^\star(c_i)\hat{r}_i + e_i$, where, given $\hat{r}_1, \ldots, \hat{r}_n$, the errors have mean zero and variance σ^2 and are uncorrelated, the expected excess risk is $(\sigma^2/n)\sum_c W(c)/\hat{W}(c)$ with realised weights $\hat{W}(c) = n^{-1}\sum_{i\in c}\hat{r}_i^2$; it equals $\sigma^2|\mathcal{P}|/n$ when $\hat{W} = W$. (b) Asymptotic: for stationary data with summable dependence and $np(c) \to \infty$ in every cell, the excess risk is $V(\mathcal{P})/n + o(1/n)$ in the limiting distribution, with $V(\mathcal{P}) = \sum_c \Omega_c/\mathbb{E}[\hat{r}^2 \mid c]$ and Ω_c the long-run variance of $\hat{r}e$ within cell c ($\Omega_c = \mathbb{E}[\hat{r}^2 e^2 \mid c]$ for independent data). (c) If every oracle gate lies in $[0, 1.5]$, clipping the estimates to $[0, 1.5]$ cannot increase the excess risk.*

Proposition 4 (Persistence-adjusted crossing condition; exact under (A1)–(A2))**.** *Let $\hat{g}_{\mathcal{P}}$ and $\hat{g}_{\mathcal{P}'}$ be unclipped least-squares gates fitted on the same sample from* V*, and write $\hat{h} = \hat{g}_{\mathcal{P}'} - \hat{g}_{\mathcal{P}} = h_{\mathrm{V}} + \varepsilon$, with $h_{\mathrm{V}} = h_{\mathrm{V}}$ the heterogeneity during gate fitting and ε its estimation error; let $h_{\mathrm{T}} = h_{\mathrm{T}}$ and take inner products under W_{T}. Assume (A1) ε has mean zero and is independent of the deployment data, and (A2) within every coarse cell the deployment weights $W_{\mathrm{T}}(c')$ of its fine cells are proportional to the realised fitting weights $\hat{W}_{\mathrm{V}}(c')$. Then the expected deployment gain of refining is*

$$G_{\mathrm{T}} = 2\langle h_{\mathrm{V}}, h_{\mathrm{T}}\rangle - \|h_{\mathrm{V}}\|^2 - \mathbb{E}\|\varepsilon\|^2,$$

and with persistence $\pi = \langle h_{\mathrm{V}}, h_{\mathrm{T}}\rangle / \|h_{\mathrm{V}}\|^2$ refining pays if and only if $\|h_{\mathrm{V}}\|^2(2\pi - 1) > \mathbb{E}\|\varepsilon\|^2$. When fitting and deployment weights coincide, $\mathbb{E}\|\varepsilon\|^2 = \Delta V/n + o(1/n)$ with $\Delta V = V(\mathcal{P}') - V(\mathcal{P})$ (Proposition 3). Without (A2) the gain has the extra term $2\mathbb{E}\langle \hat{h}, g^{\star}_{\mathcal{P}',\mathrm{T}} - \hat{g}_{\mathcal{P}}\rangle$; if ε carries the $O(1/n)$ bias of ratio estimators, G_{T} changes by $2\langle \mathbb{E}\varepsilon, h_{\mathrm{T}} - h_{\mathrm{V}}\rangle$, which vanishes when the heterogeneity persists exactly.

Proposition 5 (Tracking trade-off; exact under (A4))**.** *(A4) Let the oracle gate of a cell follow a random walk $g_t = g_{t-1} + \eta_t$ with independent increments of variance σ_η^2, let the daily estimates be $\hat{g}_t = g_t + \nu_t$ with independent noise of variance s^2, independent of the increments, and let the gate used on day t use the estimates up to day $t - \ell$, $\ell \geq 1$. The mean of the last w estimates has mean squared error*

$$\mathrm{MSE}(w) = \frac{s^2}{w} + \sigma_\eta^2\Big(\ell + \frac{(w-1)(2w-1)}{6w}\Big),$$

minimised over $w \geq 1$ at $w^\star = (3s^2/\sigma_\eta^2 + 1/2)^{1/2}$ whatever the lag [53]. Exponential forgetting with daily factor λ has $\mathrm{MSE}(\lambda) = s^2(1-\lambda)/(1+\lambda) + \sigma_\eta^2\{\ell + \lambda^2/(1-\lambda^2)\}$, and without drift ($\sigma_\eta = 0$) the expanding window is best.

In forecasting terms: structure that does not persist ($\pi \leq 1/2$), such as a zone-specific bias of one winter, never pays; persistent structure pays once there are enough observations per estimated weight; a stronger base forecast leaves less correctable variation; and updating a gate adds observations at no estimation price, whereas discounting old errors trades estimation noise for tracking. For reserve quantiles the estimation cost of a cell quantile is an expected excess pinball loss of about $\tau(1-\tau)/(2n_c f)$ at level τ, with n_c the cell's observations and f the error density at the quantile [55], so tail quantiles are expensive to condition finely.

Status of the assumptions in this study. (A0) holds by construction: a cell with fewer than 30 observations takes its parent's gate, so the propositions apply to the effective partition. (A1) holds up to serial dependence across the boundary between the gate-fitting and the deployment periods, which are disjoint in time and both separate from the corrector's training period. (A2) holds approximately when the seasonal profile of the correction energy is stable; it is the assumption that a change of season between fitting and deployment weakens. (A4) is a working model for choosing a window, not a claim about the data. Clipping is covered by Proposition 3(c), and the selection rules below use unclipped gates.

3.3. Choosing the design from validation data

Two selection rules estimate Proposition 4 from validation data only. The persistence-adjusted refinement test (PART) fits the gates on disjoint interleaved weekly blocks of the validation period; the mean cross-block product $\hat{C}$ of the unclipped block heterogeneity estimates the persistent heterogeneity, and the predicted gain is $\hat{G} = 2\hat{C} - \|\hat{h}_{\mathrm{full}}\|^2$, which needs no separate variance estimate. Formally, the gate-fitting days are split into K interleaved blocks by week, $b(i) = \lfloor (d_i - d_0)/7 \rfloor \bmod K$ ($K = 6$ for partitions, $K = 4$ for the per-instance gate). In block b (or the full set) let $\hat{g}_c^{(b)} = \sum_{i \in b \cap c} R_i \hat{r}_i / \sum_{i \in b \cap c} \hat{r}_i^2$ be the unclipped gate of cell c and $\hat{h}^{(b)}(c') = \hat{g}_{c'}^{(b)} - \hat{g}_{\mathrm{pa}(c')}^{(b)}$ the refinement of fine cell c' over its parent. With $w_{c'} = n^{-1}\sum_{i \in c'} \hat{r}_i^2$,

$$\hat{C} = \frac{1}{K(K-1)} \sum_{b \neq b'} \sum_{c'} w_{c'} \hat{h}^{(b)}(c') \hat{h}^{(b')}(c'), \qquad \hat{G} = 2\hat{C} - \sum_{c'} w_{c'} \big(\hat{h}^{(\mathrm{full})}(c')\big)^2, \tag{1}$$

where cells with fewer than $\max(5, \lfloor 30/K \rfloor)$ observations in a block (30 in the full set) are omitted.

Proposition 6 (What PART estimates; exact under (A1)–(A3))**.** *Let the block refinements be $\hat{h}^{(b)} = h_b + \nu_b$ with block noises of mean zero that are uncorrelated across blocks, let the weights $w_{c'}$ in Eq. (1) equal the deployment weights,*

and let $\bar{C} = \{K(K-1)\}^{-1} \sum_{b \neq b'} \langle h_b, h_{b'} \rangle$. *Then* $\mathbb{E}\hat{C} = \bar{C}$ *and* $\mathbb{E}\hat{G} = 2\bar{C} - \|h_{\mathrm{V}}\|^2 - \mathbb{E}\|\varepsilon\|^2$. *If (A3) the heterogeneity is as persistent between the interleaved validation blocks as between validation and deployment,* $\bar{C} = \langle h_{\mathrm{V}}, h_{\mathrm{T}} \rangle$, *then* $\mathbb{E}\hat{G} = G_{\mathrm{T}}$ *of Proposition 4.*

(A3) is the assumption on which the rule stands or falls: interleaved weekly blocks measure how persistent the heterogeneity is within the validation period, not across a change of season between validation and deployment. Clipping the block gates would bias $\hat{C}$ towards zero near the clip bounds, which is why unclipped block estimates are used. PART refines from one global gate to one gate per zone, then to the best state partition and then to the per-instance gate while $\hat{G} > 0$. Prequential validation [56] replays each time policy p day by day after a 60-day burn-in and estimates $\hat{G}_{\mathrm{pq}} = N^{-1} \sum_i [(R_i - g^{\exp}_{D_i} \hat{r}_i)^2 - (R_i - g^{p}_{D_i} \hat{r}_i)^2]$, where both gates use errors up to day $D_i - 2$ and are clipped to $[0, 1.5]$; the policy is adopted if $\hat{G}_{\mathrm{pq}} > 0$. Prequential validation needs no block structure, but it assumes, like (A3), that the drift seen during validation is representative of deployment. Both rules are compared with nested hold-out selection (fit on the first half of the validation period, score on the second) and with fixed defaults. In the State-panel study the designs are separately trained networks (Section 4); there PART is applied to the fixed-gate network's total correction, and the rule refines only if the 10th percentile of $\hat{G}$ over 50 bootstrap samples, each drawing control areas with replacement and keeping all their calibration-year days, is above zero. For the online gate, $\hat{G}$ is the prequential gain of a 180-day rolling gate over an expanding gate.

3.4. Probabilistic forecasts and reserve dimensioning

For each point forecast F the upward error quantiles of $e = Y - F$ are estimated by split-conformal empirical quantiles of out-of-sample validation errors [34] at seven levels of adaptivity: globally, per zone, per zone and 4-hour block, per zone and hour, per zone and month, per observation (gradient-boosted quantile regression) and on a rolling 90-day window updated online. Upward operating reserve is dimensioned ex ante as the 99 % quantile and evaluated by realised reliability $\Pr[e \leq \text{reserve}]$, shortfall energy $\sum \max(0, e - \text{reserve})$ and pinball loss; probabilistic accuracy is summarised by the continuous ranked probability score (CRPS) over 19 quantile levels [40].

3.5. Energy and market valuation

Three quantities are kept apart. Forecast-error energy, E_{dev} below, is the absolute error of a forecast summed over hours and annualised; it is what this study measures. Deviation energy in the regulatory sense is the gap between a State's schedule and its actual drawal, which is settled under the Deviation Settlement Mechanism [2]; the two coincide only if the schedule equals the forecast, which the States' published schedules do not (Section 5). Money values are computed under stated, stylised settlement rules, never as realised payments. The pre-registration of the hourly study names E_{dev} deviation energy; its formula is kept. For the all-India hourly series, with test hours $\mathcal{T}$, test days $\mathcal{D}$ and $\Delta t = 1\,\mathrm{h}$,

$$E_{\mathrm{dev}}(F) = \frac{365}{|\mathcal{D}|} \sum_{t \in \mathcal{T}} |Y_t - F_t| \, \Delta t, \tag{2}$$

$$\bar{\rho}(F) = |\mathcal{T}|^{-1} \sum_t \max\{0, \hat{Q}_{0.99}(F; s_t, h_t)\}, \qquad \mathrm{Rel}(F) = |\mathcal{T}|^{-1} \sum_t \mathbb{1}\{Y_t - F_t \leq \hat{Q}_{0.99}(F; s_t, h_t)\}, \tag{3}$$

$$S_{\mathrm{imb}} = \frac{365}{|\mathcal{D}|} \sum_t \big[(Y_t - B_t) - (Y_t - F_t)\big] \big(P^{\mathrm{RTM}}_t - P^{\mathrm{DAM}}_t\big) \Delta t, \tag{4}$$

where $\hat{Q}_{0.99}(F; s, h)$ is the split-conformal 99 % quantile of calibration errors of F in cell (series, hour), P^{DAM} and P^{RTM} are hourly national market clearing prices, and hours without prices contribute zero. The pre-registered reserve test requires $\bar{\rho}(B) > \bar{\rho}(F)$ and $|\mathrm{Rel}(B) - \mathrm{Rel}(F)| \leq 0.005$, a tolerance on the difference between the two forecasts, not on the nominal 0.99. Because $\hat{Q}$ is fixed before the test, $\bar{\rho}(B) - \bar{\rho}(F)$ has no sampling variation in the test period. On the State panel the forecast-error energy of design k is $E_k = \frac{365.25}{|D_k|} \sum_d \sum_{a \in \mathcal{A}_k} |y_{a,d} - \hat{y}^{0.5}_{k,a,d}|$ (GWh per year; y in GWh per day), and squared-error gains are scaled by the training-period mean absolute error of the seasonal naive forecast of each area and horizon. The comparison systems are valued in the same way with zonal prices (Supplementary Section S3). All money values are stylised: the Indian deviation charge depends on frequency and volume limits, and no redispatch, bidding response or reserve co-optimisation is modelled.

3.6. Study design and statistical analysis

All methods were developed on development data only. Three evaluations were pre-registered separately; in each, the code, hyperparameters, hypotheses and decision rules were frozen, their SHA-256 digest and a version-control

Table 2
Evaluation systems, forecasts and splits. No forecast, correction or score was computed on evaluation data before the respective pre-registration was frozen (for what had been seen, see Section 3); development data were used to design the methods.

System	Units	Forecasts corrected	Train / validation / test	Role
Indian State control areas	34 areas	gated context correction, 5 targets × 3 horizons	≤Mar 2024 / FY2024–25 / Apr 2025–Aug 2026	evaluation (own pre-registration)
All-India hourly (Grid-India SCADA)	4 series	2 naive, 6 learned	<Sep 2025 / Sep 2025–Feb 2026 / Mar–Sep 2026	evaluation (own pre-registration)
NYISO zonal load, prices, reserve prices	11 zones	ISO day-ahead; 8 learned and naive	≤2024 / 2025 / Jan–Aug 2026	comparison
OPSD / ENTSO-E load	46 areas	TSO day-ahead; 2 naive	2015–17 / 2018 / 2019	comparison
OPSD / ENTSO-E load	46 areas	TSO day-ahead; 2 naive	2015–18 / 2019 / Jan–Sep 2020	comparison
UCI electricity clients	60 clients	2 naive	2012 / 2013 / 2014	comparison
Semi-synthetic grid	12 areas	known gate structure	OPSD 2018 / 2019	comparison
NYISO zonal load; earlier India panel; ETT	–	–	up to 2025	development

tag were created before any forecast, correction or score was computed on the evaluation period, and the data-loading code refused evaluation data unless both matched (Supplementary Section S1). The State-panel test (3 checks) belongs to the pre-registration of the Indian State-day study, frozen on 18 September 2026; the all-India hourly study has its own, frozen on 6 October 2026 (hashes and tags in Supplementary Table S11), with 20 tests in seven families (simulation, mechanism, forecast strength, selection, time, uncertainty and energy) and one estimate. Before it was frozen, the hourly series had been built and summarised over the whole period and the State-panel results for an overlapping period were known, but no forecast, correction or score had been computed on the hourly test months; Supplementary Table S11 lists what had been seen before each freeze, every logged deviation and every run. The comparison study on NYISO, European and client data has eight hypotheses. Each evaluation unit (a system and test period) contributes daily totals of loss differences; 95 % confidence intervals and one-sided p-values come from a moving-block bootstrap over days [57] with 7-day blocks and 2 000 resamples, which resamples days with all series or areas together (Supplementary Table S12 lists what each analysis resamples). Effects are pooled across units by random-effects meta-analysis [58]; Holm's procedure [59] controls the error rate within each family of hypotheses and the Benjamini–Yekutieli procedure [60] across families. Gates and correctors are fitted with three seeds and the losses averaged; the learned base forecasts of the hourly study come from one seed. Analyses added after the scoring are labelled additional (post hoc) or exploratory; the exploratory checks of Section 5.3 follow a protocol whose settings were committed before they were run (file `02_design/11_ecm_revision_exploratory.md` of the replication package).

4. Case studies and data

Indian State control areas. The first case is the Indian power system, studied with its own forecaster and its own pre-registration. The data are a State-day panel of 34 Indian State control areas (all 28 States, five Union-Territory control areas and the Damodar Valley Corporation) built only from official Grid-India, CEA and IMD reports of the Government of India's grid operator, energy authority and weather service [61–64], published as an open dataset [65]. The forecaster is a neural network with a base forecast from each target's own history and one correction per block of context variables, each behind a gate $g = 1.5\,\sigma(\theta)$ learned jointly with the corrections on the fitting years, $\hat{y} = b + \sum_k g_k r_k$, for five day-ahead targets (energy met, actual drawal, conventional and renewable generation, day-ahead price) at horizons of one to three days; the price is forecast per exchange bid area, the other targets per control area. Three refinements of the fixed gates are compared: a gate per block and regime (4 seasons × 2 volatility states), trained as a separate network; a gate per instance, also a separate network; and an online gate per control area and horizon, a least-squares weight on the fixed design's total correction re-estimated each day on the last 180 days of published errors and clipped to [0, 1.5]. PART for the two partitions and prequential validation for the time policy are computed on the calibration year (FY2024–25) only, and their predicted signs are compared with the realised gains on 1 April 2025 to 31 August 2026, which were opened once, after that study's pre-registration was frozen. Gains in that test are mean reductions in squared error in units scaled by the training-period seasonal naive; the test is a one-sided binomial test of sign agreement over the 5×3 target–horizon cells, Holm-corrected over the three refinements. After the State-panel pre-registration had been scored, and before the hourly study was frozen, we added one analysis in energy units: for each design, the forecast-error energy is the absolute error of the daily median forecast summed over all control areas, in GWh per year (for the price, the mean absolute error in Rs/MWh), and every refinement is compared with the fixed gate by a 7-day block bootstrap. It uses the stored forecasts only; nothing is refitted.

Table 3
Forecast-error energy of the gate designs on Indian State control areas (energy met and drawal: 34 areas; conventional generation: 27; renewable generation: 16 areas, 1 April to 18 November 2025, where the CEA source ends; price: 13 bid areas), one day ahead, evaluation period 1 April 2025 to 31 August 2026 (additional analysis, not pre-registered). Error energy is the sum over areas of the absolute error of the daily median forecast, in GWh per year. Columns 4–6: change relative to one fixed gate, with a 7-day block-bootstrap 95 % interval; a positive change means more error.

Target	No correction	Saved by one fixed gate	Gate per regime	Gate per instance	Gate online over time
Energy met	98,346	22,911 [19,275, 27,109]	−15 [−216, +194]	+287 [−167, +668]	+189 [−200, +563]
Actual drawal	86,318	13,374 [10,273, 17,003]	−1 [−133, +123]	+39 [−221, +269]	+624 [+260, +1,035]
Conventional generation	106,128	14,601 [10,741, 18,865]	−289 [−488, −103]	+180 [−232, +614]	−56 [−546, +416]
Renewable generation	54,691	5,425 [1,392, 9,578]	+465 [+308, +601]	+342 [−87, +749]	−105 [−1,344, +1,070]
Day-ahead price[a]	1,008	1,485	1,503	1,542	956

[a] Day-ahead price: mean absolute error in Rs/MWh of each design (no correction, fixed gate, then the three refinements), per exchange bid area.

All-India hourly series. The second Indian case uses the 15-minute SCADA values that Grid-India prints in the *TimeSeries* sheet of its Daily Power Supply Position report [61] from November 2024 to 13 September 2026, averaged to hours (a value flagged as missing, out of range or frozen telemetry is left out, and an hour needs at least three valid blocks). Four series are corrected: all-India demand met, net demand met (demand met minus wind and solar), wind and solar generation, in MW. Eight day-ahead forecasts are corrected: a 168-hour seasonal naive and a 48-hour lag forecast, LightGBM [28], NHITS [22], PatchTST [23], TiDE [24], DLinear [25] and the zero-shot Chronos-Bolt [26]. For a target day every forecast uses data up to the end of the day two days before, because a day's report is published the next day; the learned models are refitted twice so that all their forecasts are out of sample. The corrector is trained on data before 1 September 2025 (for the learned forecasts, May to August 2025, the first months with out-of-sample forecasts); gates are fitted and designs selected on 1 September 2025 to 28 February 2026 and tested on 1 March to 13 September 2026. The learned base forecasts are not refitted after 1 September 2025. The corrector, gate ladder and selection rules are those of Section 3, and the semi-synthetic test of the condition is rebuilt on the daily drawal of the 34 State control areas. Imbalance cost is valued with hourly means of the national day-ahead and real-time market clearing prices in Grid-India's deviation settlement files [66], available to 6 September 2026. From 1 June 2026 the source reports storage separately; the four series keep their meaning and the change is reported, not corrected.

Comparison systems. The method was first tested, under an earlier pre-registration, outside India; those systems serve here only for comparison (Table 2). They are hourly NYISO zonal load with the ISO's day-ahead forecast and zonal prices [67], the day-ahead forecasts of 46 European TSO areas [68], whose implausible values are handled by a logged sensitivity analysis (Supplementary Section S2.1), 60 Portuguese electricity clients [69], and a semi-synthetic grid that combines real features, a real corrector and real error scales with gate heterogeneity of controlled size and persistence.

5. Results

All pre-registered results were produced once by the frozen code. Three pre-registrations apply: the Indian State-panel test, with 3 checks (Table 4); the all-India hourly study, with 20 tests (Table 5); and the earlier comparison study, in which 19 of 23 tests were supported (Supplementary Table S2). The energy valuation of the State-panel designs is an additional analysis.

5.1. Indian State control areas: forecast-error energy of each gate design

One fixed gate on the context correction paid on every energy target (Table 3). One day ahead it removed 13,374 GWh per year of forecast-error energy from drawal (15 %), 14,601 GWh from conventional generation (14 %), 22,911 GWh from demand met (23 %) and 5,425 GWh from renewable generation (10 %; 16 areas, April to November 2025, annualised), with savings of similar size two and three days ahead (Supplementary Table S4). Refining the gate rarely added to this: over the 36 combinations of energy target, horizon and refinement, a finer gate left significantly more error than the fixed gate in 9 and less in 3 (Fig. 2). The online gate per area added +624 GWh per year to drawal

Table 4
Replication on Indian State control areas (separately pre-registered, family F5). PART is computed on the calibration year and compared with the realised gain on the evaluation period, 1 April 2025 to 31 August 2026, over five targets and three horizons. Gains are mean reductions in squared error in scaled units; the rule refines only when the lower end of PART's 80 % interval is above zero. Online: one gate per area and horizon re-estimated daily on a 180-day window; for it the validation criterion is prequential validation, not PART. Rule refines: cells where the rule refined. Rule gain and Always-refine gain: mean realised gain over the 15 cells when refining only where the rule chose, or everywhere (scaled squared error).

Refinement	Validation criterion predicts a gain	Sign agreement	p_{Holm}	Verdict	Rule refines	Rule gain	Always-refine gain
instance vs. fixed	0/15	14/15	0.00146	supported	0	+0.000	-0.053
Online vs. fixed	6/15	10/15	0.302	not supported	3	+0.409	+0.062
regime vs. fixed	7/15	4/15	0.982	not supported	3	-0.011	-0.010

one day ahead (+260 to +1,035) and a gate per regime +465 GWh to renewable generation (+308 to +601), whereas a gate per regime changed the error energy of conventional generation by −289 GWh per year (−488 to −103).

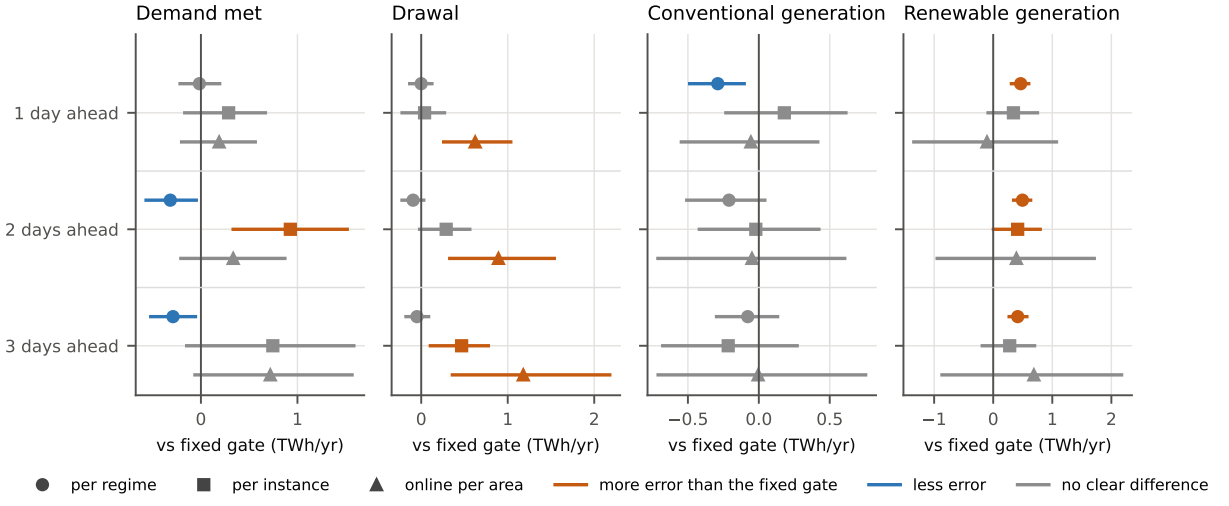


Figure 2: Indian State control areas, evaluation period 1 April 2025 to 31 August 2026 (additional analysis, not pre-registered): forecast-error energy of each finer gate design minus that of one fixed gate, summed over areas, in TWh per year, with 7-day block-bootstrap 95 % intervals, for one to three days ahead (renewable generation: 16 areas, 1 April to 18 November 2025). Positive values mean more error than the fixed gate; 9 of the 36 cells are significantly worse and 3 significantly better.

The day-ahead price behaved differently: the fixed gate, fitted on earlier years, made the forecast worse than no correction (mean absolute error 1,485 against 1,008 Rs/MWh), and re-estimating the gate online repaired this and beat both (956 Rs/MWh), as the criterion expects when the relation between context and price changes between fitting and operation.

Could the operator have known in advance? PART, computed on the calibration year only, predicted no gain from per-instance gates in all 15 cells and was right in 14 of them ($p_{\text{Holm}} = 0.00146$; Table 4): refining to instances lost accuracy on average (-0.053 in scaled squared error). For per-regime gates and adaptation over time its predicted sign was right in only 4 of 15 and 10 of 15 cells, no better than chance, so 1 of the 3 checks is supported. As a decision rule, refining only when the lower end of its 80 % bootstrap interval is above zero, it never refined to instances and adapted over time only for the day-ahead price at all three horizons, where adaptation paid (1.71 to 2.26 in scaled squared error); but it also chose 3 regime refinements, and each of them lost (Supplementary Table S4).

The States' own schedules (exploratory, after scoring). The forecasts corrected above are our own. Applied to the States' recorded drawal schedules, with the schedule published two days earlier as the forecast, one gate per area removed 5,185 GWh per year (6.3 %; 4,172 to 6,198) of the 82,835 GWh of deviation energy; of 11 finer or time-adaptive designs, 1 did better (by 259 GWh per year) and 1 worse (+1,787 GWh per year). Even the final, revised schedule carried a correctable error (28.3 % of its deviation energy), which points to persistent schedule bias.

5.2. The all-India hourly series: a second pre-registered test

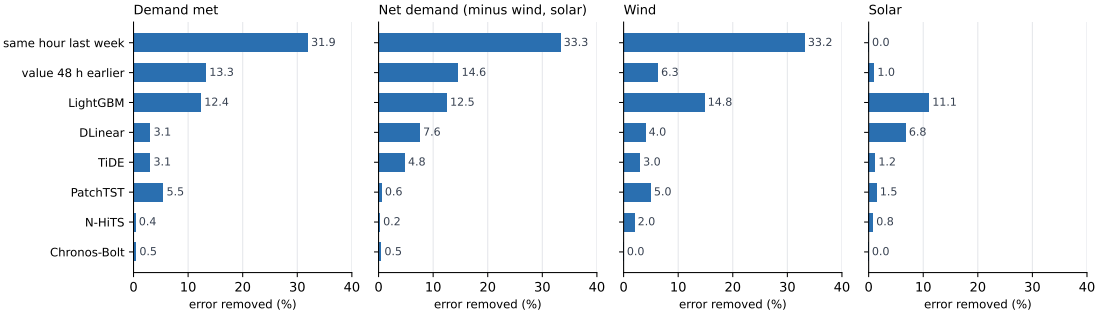


Figure 3: All-India hourly series, test months 1 March to 13 September 2026: share of squared error removed by the best of 13 gate designs, identified on the test months (an upper bound for any rule that chooses before the test), for each of the eight day-ahead forecasts and four series; each design is shown in Supplementary Table S5.

On the hourly series, 5 of the 20 pre-registered tests were supported (Table 5). How much a correction helped depended on the forecast it corrected (Fig. 3): the best of the 13 designs, identified with hindsight, removed 31.9 % of the squared error of the weekly persistence forecast of demand and 11.1 to 14.8 % from LightGBM, but at most 2.0 % from N-HiTS and Chronos-Bolt. For weekly persistence the largest gains came from gates that adapt over time or per instance; a per-series gate updated daily removed 9.7 to 28.2 % for weekly persistence and LightGBM, whereas the static per-series gate removed 8.9 % for persistence demand and added error for wind (−12.5 %; Supplementary Table S5). The pre-registered test that the best gain rises with the forecast's error across the 32 forecast–series pairs was not supported (rank correlation 0.13, $p = 0.24$).

The validation-only rules were not confirmed, and with four series the sign tests have limited power. PART predicted the sign of a refinement's gain in 0.625 of 32 cases ($p_{\text{Holm}} = 0.43$), with no lower regret than never refining, nested hold-out or fixed per-series gates, and prequential validation in only 0.425 of 40. On the semi-synthetic grid rebuilt on the 34 State control areas the crossing condition predicted the sign of partition gains in 0.718 of non-boundary configurations and PART in 0.746, short of the pre-registered thresholds. A post-hoc look (Fig. 4) shows why: with 0.25 of the gate-fitting days the refined state cells fell below the minimum cell size and took their parents' gates, so the realised gain was exactly zero in all 144 such configurations, while in the remaining 126 decisive configurations the condition predicted the sign in 99 % of cases (unit axis: 0.94). The mechanism test was reversed (more fitting data narrowed the advantage of per-instance gates, −0.029, −0.043 to −0.015); one result held: updating the gate daily was not worse than freezing it for 7 of 8 forecasts.

For all-India demand, the per-series correction of the LightGBM forecast reduced forecast-error energy (the pre-registered "deviation energy") by 5,742 GWh per year (3,601 to 7,610) and the upward reserve by 805 MW with series×hour 99 % quantiles. Both pre-registered tests were supported, but the reserve figure is not a usable reserve level: its reliability rule compares the two forecasts with each other, the difference has no sampling variation, and the quantiles, calibrated on September to February, covered only 0.717 (uncorrected) and 0.718 (corrected) of the test hours. Series×hour cells had lower 99 % pinball loss than one global quantile (by 264 MW, 149 to 370) and than per-series quantiles (by 171 MW), yet covered only 0.871, and per-instance quantiles 0.734. Valued with national day-ahead and real-time prices, the correction raised the stylised two-settlement imbalance cost by 469 Rs crore per year (188 to 776; an estimate, not a test; days after 6 September 2026 have no prices and enter as zero).

Table 5
Pre-registered tests of the all-India hourly study (test months 1 March – 13 September 2026). Holm-adjusted p within each family; threshold tests have no p-value. Energy rows refer to all-India demand met and the LightGBM forecast. The reserve row compares quantile tables fixed before the test, so its bootstrap interval is degenerate; its reliability rule compares corrected with uncorrected coverage within ± 0.005 (0.718 against 0.717; nominal 0.99).

Family	Test	Estimate	p_{Holm}	Verdict
Synthetic	Crossing condition predicts partition gains (oracle, ≥0.85)	0.718	–	not supported
Synthetic	PART predicts partition gains (≥0.75)	0.746	–	not supported
Synthetic	Crossing condition predicts per-instance gains (oracle, ≥0.85)	0.825	–	not supported
Synthetic	PART predicts per-instance gains (≥0.75)	0.375	–	not supported
Mechanism	More fitting data helps the per-instance gate	−0.029	1.00	not supported
Baseline	Weaker forecasts gain more (rank correlation)	0.132	0.24	not supported
Selection	PART predicts the sign of refinement gains	0.625	0.43	not supported
Selection	PART regret below never refining	−0.001	0.21	not supported
Selection	PART regret below nested hold-out	−0.00005	1.00	not supported
Selection	PART static choice below nested hold-out	0.002	1.00	not supported
Selection	PART static choice below per-series gates	−0.004	0.43	not supported
Time	Prequential validation predicts discounting gains	0.425	1.00	not supported
Time	Prequential regret below always discounting	0.002	1.00	not supported
Time	Prequential regret below never discounting	−0.004	0.77	not supported
Time	Updating the gate daily costs nothing (≥80 %)	0.875	–	supported
Probabilistic	Series×hour quantiles beat one global quantile (pinball, MW)	264	<0.01	supported
Probabilistic	Series×hour quantiles beat per-series quantiles (pinball, MW)	171	<0.01	supported
Probabilistic	Per-instance quantiles under-cover, cells do not	0.734	–	not supported
Energy	Correction lowers demand deviation energy (GWh/yr)	5,742	<0.01	supported
Energy	Correction lowers 99 % upward reserve (MW)	805	<0.01	supported
Estimate	Imbalance cost saved (Rs crore/yr; estimate only)	−469	–	estimate

Table 6
All-India hourly series, test months 1 March – 13 September 2026: share of the squared error of each day-ahead forecast removed by each correction, averaged over the four series (%). [P] pre-registered design; [H] best of the 13 gate designs, chosen with hindsight on the test months; [E] exploratory, added after scoring. Conventional: the bias correction, refitted daily on the forecast's own errors up to day $D-2$, that had the lowest error on the gate-fitting months. Negative values add error.

	Learned gated correction			Using the forecast's own errors		
Forecast	Static [P]	Daily [P]	Best [H]	Gated [E]	Conventional [E]	Both [E]
Same hour last week	4.3	14.9	24.6	31.9	39.7	41.7
Value 48 h earlier	1.5	5.7	8.8	10.4	10.8	13.2
LightGBM	7.1	11.5	12.7	12.8	34.7	34.5
Chronos-Bolt	−1.6	−0.4	0.2	0.0	−0.2	0.3
N-HiTS	−0.8	0.0	0.8	0.9	−4.1	−5.3
PatchTST	1.3	1.2	3.1	5.1	13.7	13.7
TiDE	1.0	0.7	3.0	3.0	5.2	5.0
DLinear	1.2	1.2	5.4	8.6	18.4	18.4

Static: one gate per series; Daily: the per-series gate updated with each day's errors; Gated [E]: best design of the corrector that also sees the forecast and its lagged errors; Both: the gated correction applied on top of the conventional one.

5.3. Benchmarks and robustness checks on the Indian data (exploratory)

These checks were added after both pre-registrations had been scored, with every setting fixed in a protocol that was committed before they were run; they are exploratory, and none changes a pre-registered verdict.

Does a conventional bias correction do better? Table 6 compares the gated corrections with the bias corrections that operators use [19, 20]: hour-of-day and hour-of-week means of the forecast's own recent errors, the latter with an

autoregressive remainder, and a linear regression on lagged errors, each refitted daily on the errors published up to day $D-2$ and chosen per series on the gate-fitting months. The conventional correction removed 39.7 % of the squared error of weekly persistence, 34.7 % of LightGBM's and 18.4 % of DLinear's, against at most 12.7 % for LightGBM from any gated design chosen with hindsight. It was significantly better than the daily-updated gate for 4 of the 8 forecasts and significantly worse for 0. Giving the learned corrector the forecast and its lagged errors raised its gains (the per-series gate on weekly persistence went from 4.3 to 30.0 %) but did not close the gap, and a gated correction applied on top of the conventional one added at most 2.3 percentage points. Combining all gate designs online by Fixed Share aggregation [54] did not change this (mean skill 0.069, against 0.043 for the daily-updated gate), and no correction of either kind helped N-HiTS or Chronos-Bolt. A bias that drifts with the seasons is tracked better by a simple model refitted on recent errors than by a rich correction trained once: this is the tracking trade-off of Proposition 5 applied to the whole correction, not only to its gate.

Is the energy gain an artefact of a stale forecast? Refitting the LightGBM forecast at the start of every month reduced its average under-forecast of all-India demand over the test months from 11,963 to 4,172 MW. The static correction of the refreshed forecast still lowered forecast-error energy by 3,574 GWh per year (741 to 6,658), against 5,742 GWh for the forecast that was not refitted, and it again raised the stylised imbalance cost (−510 Rs crore per year, −1,019 to −32).

Did the seasonal split defeat the selection rules? The validation months (September to February) and the test months (March to September) lie in different seasons, which could break assumption (A3). We therefore refitted the gates and recomputed both rules at the start of every test month on the 182 days ending two days earlier. The corrector refitted for this check reproduced the stored corrections to within 1.5×10^{-11} MW. With these rolling windows PART predicted the sign of the monthly gain of a partition refinement in 0.63 of 168 cases, against 0.59 with the fixed validation window, and prequential validation in 0.43 against 0.42 of 280. Moving the validation window up to the month of use therefore helped little: on these series the variation that finer or faster gates exploit does not persist even from one month to the next.

Reserve quantiles in the season of use. The ex-ante series×hour 99 % quantiles of the LightGBM demand forecast covered 0.98 of the hours in March and as little as 0.39 in the worst month, whereas the pre-registered quantile updated online over the last 90 days kept coverage between 0.88 and 1.00 in every month (Supplementary Table S14). Tracking the quantile online with the two-day publication lag [38], with the step size chosen on the development months, raised coverage from 0.717 to 0.981 and cut the shortfall energy from 23,351 to 600 GWh per year, but it needed 40,529 MW of upward reserve on average instead of 18,391 MW. A reliable reserve for this forecast is about twice what the fixed table implies.

How the settlement rule changes the value. The correction raised day-ahead purchases of demand by 1,013 MW on average while the real-time price averaged −638 Rs/MWh relative to the day-ahead price, which alone accounts for −565 Rs crore per year of the stylised cost change. Valuing the deviation energy at the day-ahead price instead gives a saving of 2,913 Rs crore per year (1,689 to 3,924), and a DSM-style buyer rule that charges over-drawal at the higher of the two market prices and credits under-drawal at 90 % of it gives 186 Rs crore (100 to 264). None of these is the settlement of an actual State, which works per 15-minute block with frequency-dependent rates and volume limits [2]; the value of a correction therefore has to be judged in the settlement it feeds.

Inference. The conclusions do not depend on the bootstrap. With 7-, 14- and 28-day moving blocks and a stationary bootstrap, the interval of the deviation-energy saving stayed within 3,345 to 7,825 GWh per year and that of the imbalance change below −188 Rs crore, and on the State panel finer gates were significantly worse than the fixed gate in 9 to 11 and better in 3 to 3 of the 36 cells. Days are resampled with all series or control areas together, so dependence between areas on the same day is preserved.

5.4. Comparison with four other power systems

Under the earlier pre-registration the same method was tested on NYISO zonal load, European TSO forecasts, electricity clients and a semi-synthetic grid (details in Supplementary Section S3). There the selection rules and the mechanism behind them were confirmed: PART predicted the sign of refinement gains in 75 % of 68 refinements ($p = 2.2 \times 10^{-5}$) with lower decision regret than never refining, prequential validation predicted the sign of discounting

gains in 75 % of cases, and fitting fine gates on more data raised the per-instance gain (pooled 0.0274). One gate per NYISO zone cut the forecast-error energy of the ISO day-ahead forecast by 853.7 GWh per year (626.1 to 1053.5) without lowering its imbalance cost, and the most flexible conditional quantile model covered only 0.9694 of hours. On the European-based semi-synthetic grid the crossing condition predicted partition gains in 93 % of non-boundary configurations (0.8485 after a logged data-quality filter; Supplementary Section S2.1, Supplementary Fig. S1).

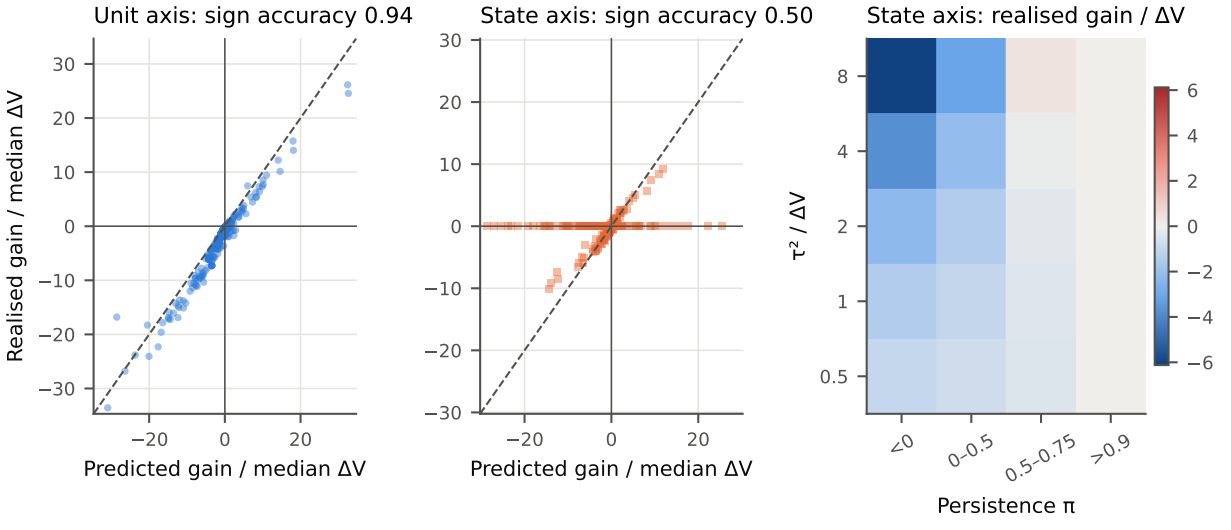


Figure 4: Pre-registered semi-synthetic test rebuilt on the daily drawal of the 34 Indian State control areas (confirmatory run; stored results). Left and centre: realised against predicted gain of refining along the unit and the state axis, scaled by the median estimation cost ΔV; the dashed line marks equality, and the title gives the share of configurations with a predicted gain of at least $\Delta V/2$ in absolute value whose sign was predicted correctly. The horizontal line of zero realised gains in the centre panel consists of the configurations fitted on a quarter of the days, whose refined cells fell back to their parents' gates. Right: mean realised gain along the state axis, relative to ΔV, by persistence π and by the injected heterogeneity relative to the estimation cost ($\tau^2/\Delta V$). Refining pays only where the heterogeneity persists ($\pi > 1/2$), as Proposition 4 states.

Set side by side, two findings carried over to India: a coarse correction lowers forecast-error energy, and lower forecast-error energy does not by itself lower imbalance cost. Two did not: on the Indian hourly series the validation-only rules no longer told when a finer gate would pay, and more fitting data did not help per-instance gates.

6. Discussion

6.1. A deployment procedure for Indian control areas

The evidence supports a simple procedure for State load despatch centres, regional and national despatch centres and forecast vendors who correct an existing day-ahead forecast.

Step 1. Start from a simple correction refitted on recent errors: a bias model of the forecast's own errors published up to day $D-2$, refitted daily. On the all-India series it beat every learned gated correction for 4 of 8 forecasts (exploratory, Table 6). Where a learned correction is used, fit one coarse gate on the most recent full year of out-of-sample errors and update it with each day's errors: on the State panel one fixed gate captured most of the gain of the designs tested, and on the hourly series updating the gate daily was not worse than freezing it for 7 of 8 forecasts.

Step 2. Expect the largest gains where the forecast is simplest; for strong learned or foundation-model forecasts no design helped, and coarse gates made them slightly worse.

Step 3. Do not refine to regime, hour or per-instance gates on the strength of a validation-only test alone: such tests did not predict which refinement would pay on the hourly series, and on the State panel they were reliable only for large, persistent changes such as the shift in the day-ahead price.

Step 4. Check reserve quantiles in the season in which they will be used.

Step 5. Judge the corrected forecast in the settlement it feeds: lower deviation energy did not lower the stylised imbalance cost in India or in New York.

6.2. Implications for Indian institutions

Every quantity in this study can be computed from reports that the Government of India's grid operator, energy authority and weather service already publish, so the checks can be repeated by the institutions that would act on them. For State, regional and national load despatch centres, which must estimate demand a day ahead and compute and publish their forecast errors [3, Regulation 31], the published errors are all that the recommended correction and its validation need; it is cheap to run and easy to audit. For the regulator, the value of a better forecast depended on the settlement rule: the same correction raised a stylised two-settlement cost and lowered a DSM-style charge (Section 5.3), so the design of deviation charges shapes the incentive to forecast well [2]. For reserve planning under Regulation 30(11) of the Grid Code [3], percentile-based requirements need quantiles recalibrated in the season of use; a fixed table covered far fewer hours than intended in the summer and monsoon months, and a recalibrated one needed about twice the reserve. Finally, the base forecasts corrected here are our own because the States' day-ahead forecasts are not published in machine-readable form; publishing them next to the actual values would let the same tests be run on the forecasts the States actually use.

6.3. Why the selection rules did not carry over

The validation-only rules were confirmed on New York, European and client data but not on the Indian hourly series. The hourly study has only four series, so its sign tests rest on few cases; its validation and test months lie in different seasons; and from June 2026 the source reports storage separately. The rolling-origin check (Section 5.3) speaks against the seasonal explanation as the main one: with gate-fitting windows ending two days before each test month the rules predicted little better than with the fixed window. What the Indian series lack is heterogeneity that persists at all: in the terms of Proposition 4 the persistence π of finer structure is low, so its sign cannot be learned from the past. A test period spanning several years would show whether this holds beyond 2026.

6.4. Implications for reserves and markets

Reserve sizing is where adaptivity is most dangerous: tail quantiles are expensive to estimate, and on both the NYISO and the Indian data the most flexible quantile model under-covered the tail. For probabilistic reserve dimensioning [29–31] this argues for conditioning on persistent structure, recalibrating online in the season of use and reporting shortfall energy alongside reserve volume. A correction that removes a bias moves purchases between market stages, so its value depends on the price spread and the settlement rule; gates fitted under a decision-aware loss [47] would target that value directly, which we did not test.

6.5. Limitations

On the State panel the data are daily, and forecast-error energy is not the settled deviation energy, because a State's schedule is not its forecast; the designs are separately trained networks, so their differences mix gate adaptivity with retraining. The hourly study covers four national series and six and a half test months of SCADA telemetry, which can contain errors and omits generation not seen by the transmission system; its corrector does not see the base forecast or its past errors, and the learned base forecasts were not refitted after August 2025. The imbalance costs are stylised two-settlement valuations, not the deviation charge of the 2024 Regulations, and the NYISO reserve valuation prices a volume that NYISO sizes by contingency rules. One corrector family with fixed hyperparameters was used, without weather forecasts, and neural forecasts were trained with a reduced number of steps fixed before any result. Validation-only rules cannot anticipate structure that changes after the validation period. Error-correction stages of the hybrid forecasting literature were not tested as baselines, and the States' own schedules and the online combination of designs were tested only in exploratory checks on periods that had already been opened. What had been seen before each freeze is listed in Section 3.

7. Conclusions

Making a forecast correction more adaptive is a bet that the structure it exploits is large, persistent and estimable. The criterion of Section 3.2 states the bet precisely, and PART estimates it from validation data exactly when the

heterogeneity is as persistent between validation weeks as between validation and use. On official Indian data the bet rarely paid. On 34 State control areas finer gates were rarely better than one fixed gate; the clear gain from adaptation over time was the day-ahead price, whose relation to the context had changed. On the all-India hourly series no gate design helped the strongest forecasts, and the validation-only rules confirmed on New York, European and client data did not tell in advance which design would pay, because finer structure did not persist even from month to month. Exploratory benchmarks added a practical lesson: a simple bias correction refitted daily on each forecast's own errors beat the learned gated corrections, and the money value of any correction depended on the settlement it feeds.

These results rest on six and a half test months, four national series and stylised valuations. Before deployment, a State or regional load despatch centre should run the same checks on its own forecasts: start from a simple correction refitted on recent errors, adopt finer or learned corrections only when their structure is shown to persist into the season of use, recalibrate reserve quantiles online, and judge every correction by the deviation settlement it serves.

A. Proofs

Cells are those of the effective partition after the minimum-cell rule, (A0) holds, and for functions constant on cells $\langle f, h\rangle = \sum_c W(c) f(c) h(c)$ under the stated weights. The proofs of the standard results, Propositions 1–3 and 5, are given in Supplementary Section S6.

Proposition 4. Condition on the fitting sample, so that $\hat{g}_{\mathcal{P}}$ and $\hat{h}$ are fixed and, by (A1), independent of the deployment data. Take $g_a = \hat{g}_{\mathcal{P}}$ and $g_b = \hat{g}_{\mathcal{P}} + \hat{h}$ in Proposition 1 under T, with $d = \hat{h}$ and $e_a = R - \hat{g}_{\mathcal{P}}\hat{r}$. For a fine cell $c' \subseteq c$, $\mathbb{E}_{\mathrm{T}}[\hat{r} e_a \mid c'] = \mathbb{E}_{\mathrm{T}}[R\hat{r} \mid c'] - \hat{g}_{\mathcal{P}}(c)\mathbb{E}_{\mathrm{T}}[\hat{r}^2 \mid c'] = \mathbb{E}_{\mathrm{T}}[\hat{r}^2 \mid c']\{h_{\mathrm{T}}(c') + g^{\star}_{\mathcal{P},\mathrm{T}}(c) - \hat{g}_{\mathcal{P}}(c)\}$, so

$$\mathbb{E}_{\mathrm{T}}[\hat{h}\hat{r}e_a] = \langle \hat{h}, h_{\mathrm{T}}\rangle + \textstyle\sum_c \{g^{\star}_{\mathcal{P},\mathrm{T}}(c) - \hat{g}_{\mathcal{P}}(c)\} \sum_{c' \subseteq c} W_{\mathrm{T}}(c')\hat{h}(c'). \tag{A.1}$$

The unclipped coarse gate is the $\hat{W}_{\mathrm{V}}$-weighted mean of the fine gates of its cell,

$$\hat{g}_{\mathcal{P}}(c) = \frac{\sum_{c' \subseteq c} \hat{W}_{\mathrm{V}}(c')\, \hat{g}_{\mathcal{P}'}(c')}{\sum_{c' \subseteq c} \hat{W}_{\mathrm{V}}(c')},$$

so $\sum_{c' \subseteq c} \hat{W}_{\mathrm{V}}(c')\hat{h}(c') = 0$; under (A2) $W_{\mathrm{T}}(c') = \kappa_c \hat{W}_{\mathrm{V}}(c')$ and the inner sum in (A.1) is zero. With $\mathbb{E}_{\mathrm{T}}[\hat{h}^2\hat{r}^2] = \|\hat{h}\|^2$, the conditional gain is $2\langle \hat{h}, h_{\mathrm{T}}\rangle - \|\hat{h}\|^2 = 2\langle h_{\mathrm{V}}, h_{\mathrm{T}}\rangle - \|h_{\mathrm{V}}\|^2 + 2\langle \varepsilon, h_{\mathrm{T}} - h_{\mathrm{V}}\rangle - \|\varepsilon\|^2$. Its expectation over the fitting sample under $\mathbb{E}\varepsilon = 0$ is G_{T}; with a bias $\mathbb{E}\varepsilon \neq 0$ the linear term contributes $2\langle \mathbb{E}\varepsilon, h_{\mathrm{T}} - h_{\mathrm{V}}\rangle$. Dividing $G_{\mathrm{T}} > 0$ by $\|h_{\mathrm{V}}\|^2$ gives the persistence form. When the realised fitting, population fitting and deployment weights coincide, $\hat{g}_{\mathcal{P}} - g^{\star}_{\mathcal{P}} = \Pi(\hat{g}_{\mathcal{P}'} - g^{\star}_{\mathcal{P}'})$, with Π the weighted projection on functions constant on $\mathcal{P}$-cells, so $\varepsilon = (I - \Pi)(\hat{g}_{\mathcal{P}'} - g^{\star}_{\mathcal{P}'})$ and, by Pythagoras, $\|\varepsilon\|^2 = \|\hat{g}_{\mathcal{P}'} - g^{\star}_{\mathcal{P}'}\|^2 - \|\hat{g}_{\mathcal{P}} - g^{\star}_{\mathcal{P}}\|^2$, whose expectation is $\Delta V/n + o(1/n)$ by Proposition 3. □

Proposition 6. The weights $w_{c'}$ are fixed by the full sample. For $b \neq b'$, $\mathbb{E}\langle \hat{h}^{(b)}, \hat{h}^{(b')}\rangle = \langle h_b, h_{b'}\rangle + \langle h_b, \mathbb{E}\nu_{b'}\rangle + \langle \mathbb{E}\nu_b, h_{b'}\rangle + \mathbb{E}\langle \nu_b, \nu_{b'}\rangle = \langle h_b, h_{b'}\rangle$, because the block noises have mean zero and are uncorrelated across blocks; averaging over the $K(K-1)$ ordered pairs gives $\mathbb{E}\hat{C} = \bar{C}$. For the full sample, $\mathbb{E}\|\hat{h}^{(\mathrm{full})}\|^2 = \|h_{\mathrm{V}}\|^2 + 2\langle h_{\mathrm{V}}, \mathbb{E}\varepsilon\rangle + \mathbb{E}\|\varepsilon\|^2 = \|h_{\mathrm{V}}\|^2 + \mathbb{E}\|\varepsilon\|^2$ under (A1). Hence $\mathbb{E}\hat{G} = 2\bar{C} - \|h_{\mathrm{V}}\|^2 - \mathbb{E}\|\varepsilon\|^2$, which equals G_{T} when (A3) gives $\bar{C} = \langle h_{\mathrm{V}}, h_{\mathrm{T}}\rangle$. Interleaving by week keeps adjacent days in different blocks only at week boundaries, so the covariance of block noises is small when the dependence of the errors decays within a week; clipped block gates would bias $\hat{C}$ towards zero near the clip bounds, which is why unclipped estimates are used. □

Data availability

The RE-FUSED replication package is openly available at `https://github.com/Ohi-Me/RE-FUSED` (folders `Forecast_Correction_India` for the hourly study and `India_Grid_Study` for the State panel; the file `FINAL_VERSION.md` maps every result to its code, result file and run log) and is archived on Zenodo as version 1.0.0, `https://doi.org/10.5281/zenodo.23301898` [65], with the data under CC BY 4.0 and the code under the MIT licence. It contains the code, the frozen pre-registrations, the development log, the processed Indian datasets including the all-India hourly series, all derived results and run logs, and SHA-256 checksums of every file. UCI electricity data (CC BY 4.0, 69) and OPSD time series (attribution required, 68) are obtained by the included download scripts, which

verify their checksums. NYISO Market Information System data [67] are public but have no redistribution licence, so the package holds the download script, the hashes of the monthly files and all derived results. The raw Grid-India, CEA and IMD report files and the development data from the India Data Portal are not redistributed; the record lists the source URL and SHA-256 hash of each report file.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Rohit Kumar: Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Investigation, Visualization, Writing – original draft. **Urvashi Bansal:** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used Claude (Anthropic) to assist with English language editing, drafting and revising text, consistency checking across tables and narrative text and formatting of the manuscript. All experimental design, data collection, model development, the original code implementation, and interpretation of results were performed entirely by the authors. The authors carefully reviewed and verified all AI-assisted output and take full responsibility for the content of this publication.

Acknowledgements

The authors acknowledge the computational facilities at Dr. B. R. Ambedkar National Institute of Technology Jalandhar, including the high-performance computing (HPC) and GPU resources used to conduct the experiments reported in this study. This research has also informed a proposal submitted under the Central Power Research Institute (CPRI) Research Scheme on Power (RSoP) programme.

References

[1] D. P. Mishra, S. Samal, R. Kumar, A. K. Sahoo, S. R. Salkuti, Renewable energy optimization for sustainable power generation, International Journal of Informatics and Communication Technology 15 (2026) 365–373. doi:10.11591/ijict.v15i1.pp365-373.

[2] Central Electricity Regulatory Commission, Central electricity regulatory commission (deviation settlement mechanism and related matters) regulations, 2024, Notification No. L-1/260/2021/CERC, dated 5 August 2024, 2024. URL: https://www.cercind.gov.in/regulations/192-Noti.pdf.

[3] Central Electricity Regulatory Commission, Central Electricity Regulatory Commission (Indian Electricity Grid Code) Regulations, 2023, Notification No. L-1/265/2022/CERC, dated 29 May 2023, 2023. URL: https://www.cercind.gov.in/Regulations/180-Regulations.pdf.

[4] Central Electricity Regulatory Commission, Notification of the date of coming into force of the CERC (deviation settlement mechanism and related matters) regulations, 2024, No. L-1/260/2021/CERC, dated 27 August 2024, 2024. URL: https://cercind.gov.in/Regulations/192-effective-160924.pdf.

[5] Press Information Bureau, Ministry of Power, Government of India, India meets all-time highest peak power demand of ∼256 GW without shortage, Release ID 2256313, 28 April 2026, 2026. URL: https://www.pib.gov.in/PressReleasePage.aspx?PRID=2256313.

[6] R. Bareth, M. Kochar, A. Yadav, M. Pazoki, Load forecasting model using LSTM for Indian state load dispatch centre, Electrica 24 (2024) 601–615. doi:10.5152/electrica.2024.23158.

[7] A. D. C. Arul Babu, V. Ramadesigan, R. Banerjee, Medium and long-term electricity demand forecasting for India – incorporating EV charging and space cooling impacts, Applied Energy 420 (2026) 128069. doi:10.1016/j.apenergy.2026.128069.

[8] A. Desai, I. Mukhopadhyay, A. Ray, Economic modeling of deviation settlement under solar forecasting and scheduling, in: 2020 47th IEEE Photovoltaic Specialists Conference (PVSC), 2020, pp. 2444–2448. doi:10.1109/PVSC45281.2020.9300572.

[9] F. Carrasco, F. Kristjanpoller, I. Verdugo, P. Viveros, R. Pascual, Forecast-informed daily energy dispatch for cost and emission reduction: Evidence from an emerging economy power system, Energy Conversion and Management 364 (2026) 121611. doi:10.1016/j.enconman.2026.121611.

[10] T. Hong, P. Pinson, Y. Wang, R. Weron, D. Yang, H. Zareipour, Energy forecasting: A review and outlook, IEEE Open Access Journal of Power and Energy 7 (2020) 376–388. doi:10.1109/oajpe.2020.3029979.

[11] F. Petropoulos, D. Apiletti, V. Assimakopoulos, M. Z. Babai, D. K. Barrow, S. Ben Taieb, C. Bergmeir, R. J. Bessa, J. Bijak, J. E. Boylan, J. Browell, C. Carnevale, J. L. Castle, P. Cirillo, M. P. Clements, C. Cordeiro, F. L. Cyrino Oliveira, S. De Baets, A. Dokumentov, J. Ellison, P. Fiszeder, P. H. Franses, D. T. Frazier, M. Gilliland, M. S. Gönül, P. Goodwin, L. Grossi, Y. Grushka-Cockayne, M. Guidolin, M. Guidolin, U. Gunter, X. Guo, R. Guseo, N. Harvey, D. F. Hendry, R. Hollyman, T. Januschowski, J. Jeon, V. R. R. Jose, Y. Kang, A. B. Koehler, S. Kolassa, N. Kourentzes, S. Leva, F. Li, K. Litsiou, S. Makridakis, G. M. Martin, A. B. Martinez, S. Meeran, T. Modis, K. Nikolopoulos, D. Önkal, A. Paccagnini, A. Panagiotelis, I. Panapakidis, J. M. Pavía, M. Pedio, D. J. Pedregal, P. Pinson, P. Ramos, D. E. Rapach, J. J. Reade, B. Rostami-Tabar, M. Rubaszek, G. Sermpinis, H. L. Shang, E. Spiliotis, A. A. Syntetos, P. D. Talagala, T. S. Talagala, L. Tashman, D. Thomakos, T. Thorarinsdottir, E. Todini, J. R. Trapero Arenas, X. Wang, R. L. Winkler, A. Yusupova, F. Ziel, Forecasting: theory and practice, International Journal of Forecasting 38 (2022) 705–871. doi:10.1016/j.ijforecast.2021.11.001.

[12] M. G. Pinheiro, S. C. Madeira, A. P. Francisco, Short-term electricity load forecasting—a systematic approach from system level to secondary substations, Applied Energy 332 (2023) 120493. doi:10.1016/j.apenergy.2022.120493.

[13] J. Xie, Y. Hua, M. Cheng, F. Salim, H. Xue, Causality-inspired safe residual correction for multivariate time series, 2025. Preprint, arXiv:2512.22428.

[14] J. Liu, X. Huang, Q. Li, Z. Chen, G. Liu, Y. Tai, Hourly stepwise forecasting for solar irradiance using integrated hybrid models CNN-LSTM-MLP combined with error correction and VMD, Energy Conversion and Management 280 (2023) 116804. doi:10.1016/j.enconman.2023.116804.

[15] J. Wang, Y. Qian, L. Zhang, K. Wang, H. Zhang, A novel wind power forecasting system integrating time series refining, nonlinear multi-objective optimized deep learning and linear error correction, Energy Conversion and Management 299 (2024) 117818. doi:10.1016/j.enconman.2023.117818.

[16] Y. Hao, X. Wang, J. Wang, W. Yang, A new perspective of wind speed forecasting: Multi-objective and model selection-based ensemble interval-valued wind speed forecasting system, Energy Conversion and Management 299 (2024) 117868. doi:10.1016/j.enconman.2023.117868.

[17] G. Casciaro, F. Ferrari, M. Cavaiola, A. Mazzino, Novel strategies of ensemble model output statistics (EMOS) for calibrating wind speed/power forecasts, Energy Conversion and Management 271 (2022) 116297. doi:10.1016/j.enconman.2022.116297.

[18] M. J. Mayer, D. Yang, D. Markovics, The complexity and dimensionality of making deterministic photovoltaic power forecasts from ensemble numerical weather prediction, Energy Conversion and Management 344 (2025) 120303. doi:10.1016/j.enconman.2025.120303.

[19] T. Möbius, M. Watermeyer, O. Grothe, F. Müsgens, Enhancing energy system models using better load forecasts, Energy Systems 16 (2023) 573–602. doi:10.1007/s12667-023-00590-3.

[20] K. Xie, A. Giacomoni, X. Deng, Y. Wu, Bias calibration and error propagation adjustment for ML-based time series forecasting: A systematic study for PJM's electricity load forecast amid Virginia's data center surge, Energy 336 (2025) 138411. doi:10.1016/j.energy.2025.138411.

[21] J. de Vilmarest, J. Browell, M. Fasiolo, Y. Goude, O. Wintenberger, Adaptive probabilistic forecasting of electricity (net-)load, IEEE Transactions on Power Systems 39 (2024) 4154–4163. doi:10.1109/TPWRS.2023.3310280.

[22] C. Challu, K. G. Olivares, B. N. Oreshkin, F. Garza Ramirez, M. Mergenthaler Canseco, A. Dubrawski, NHITS: Neural hierarchical interpolation for time series forecasting, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2023, pp. 6989–6997. doi:10.1609/aaai.v37i6.25854.

[23] Y. Nie, N. H. Nguyen, P. Sinthong, J. Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers, in: International Conference on Learning Representations, 2023. URL: https://openreview.net/forum?id=Jbdc0vTOcol.

[24] A. Das, W. Kong, A. Leach, S. Mathur, R. Sen, R. Yu, Long-term forecasting with TiDE: Time-series dense encoder, Transactions on Machine Learning Research (2023). URL: https://openreview.net/forum?id=pCbC3aQB5W.

[25] A. Zeng, M. Chen, L. Zhang, Q. Xu, Are transformers effective for time series forecasting?, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2023, pp. 11121–11128. doi:10.1609/aaai.v37i9.26317.

[26] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, Y. Wang, Chronos: Learning the language of time series, Transactions on Machine Learning Research (2024). URL: https://openreview.net/forum?id=gerNCVqqtR.

[27] S. Guarino, A. Buscemi, V. Lo Brano, Photovoltaic power forecasting and anomaly detection: a comparative study of statistical, deep learning, and zero-shot foundation models on seven years of real plant data, Energy Conversion and Management 369 (2026) 122078. doi:10.1016/j.enconman.2026.122078.

[28] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, in: Advances in Neural Information Processing Systems 30, 2017.

[29] H. Holttinen, M. Milligan, E. Ela, N. Menemenlis, J. Dobschinski, B. Rawn, R. J. Bessa, D. Flynn, E. Gomez-Lazaro, N. K. Detlefsen, Methodologies to determine operating reserves due to increased wind power, IEEE Transactions on Sustainable Energy 3 (2012) 713–723. doi:10.1109/TSTE.2012.2208207.

[30] N. Costilla-Enriquez, M. Ortega-Vazquez, A. Tuohy, A. Motley, R. Webb, Operating dynamic reserve dimensioning using probabilistic forecasts, IEEE Transactions on Power Systems 38 (2023) 603–616. doi:10.1109/TPWRS.2022.3163106.

[31] Y. Sun, J. H. Nelson, J. C. Stevens, A. H. Au, V. Venugopal, C. Gulian, et al., Machine learning derived dynamic operating reserve requirements in high-renewable power systems, Journal of Renewable and Sustainable Energy 14 (2022) 036303. doi:10.1063/5.0087144.

[32] T. Zhang, R. Shi, S. Ma, Q. Li, L. Jia, Deep generative model-based energy and reserve scheduling for frequency security management of regional integrated energy systems, Energy Conversion and Management 344 (2025) 120219. doi:10.1016/j.enconman.2025.120219.

[33] J. Zhu, Y. He, X. Yang, S. Yang, Ultra-short-term wind power probabilistic forecasting based on an evolutionary non-crossing multi-output quantile regression deep neural network, Energy Conversion and Management 301 (2024) 118062. doi:10.1016/j.enconman.2024.118062.

[34] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman, Distribution-free predictive inference for regression, Journal of the American Statistical Association 113 (2018) 1094–1111. doi:10.1080/01621459.2017.1307116.

[35] I. Gibbs, E. Candès, Adaptive conformal inference under distribution shift, in: Advances in Neural Information Processing Systems, volume 34, 2021, pp. 1660–1672.

[36] M. Zaffran, O. Féron, Y. Goude, J. Josse, A. Dieuleveut, Adaptive conformal predictions for time series, in: Proceedings of the 39th International Conference on Machine Learning, Proceedings of Machine Learning Research 162, 2022, pp. 25834–25866. URL: https://proceedings.mlr.press/v162/zaffran22a.html.

[37] R. F. Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani, Conformal prediction beyond exchangeability, The Annals of Statistics 51 (2023) 816–845. doi:10.1214/23-AOS2276.

[38] A. N. Angelopoulos, E. J. Candès, R. J. Tibshirani, Conformal PID control for time series prediction, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023), 2023. doi:10.52202/075280-1000.

[39] C. O'Connor, M. Bahloul, R. Rossi, S. Prestwich, A. Visentin, Conformal prediction for electricity price forecasting in the day-ahead and real-time balancing market, Energy and AI 21 (2025) 100571. doi:10.1016/j.egyai.2025.100571.

[40] T. Gneiting, A. E. Raftery, Strictly proper scoring rules, prediction, and estimation, Journal of the American Statistical Association 102 (2007) 359–378. doi:10.1198/016214506000001437.

[41] K. Maciejowska, A. Lipiecki, B. Uniejewski, Statistical and economic evaluation of forecasts in electricity markets: Beyond RMSE and MAE, Energy Conversion and Management 356 (2026) 121408. doi:10.1016/j.enconman.2026.121408.

[42] M. J. Mayer, D. Yang, D. Markovics, Quality and economic value of operational photovoltaic power forecasts in the day-ahead market, Energy Conversion and Management 358 (2026) 121451. doi:10.1016/j.enconman.2026.121451.

[43] M. Pierro, C. Cornaro, L. Fialho, R. Perez, J. Remund, G. Barchi, Firm PV generation through optimized trading and forecasting: Economic and system flexibility impacts, Energy Conversion and Management 368 (2026) 121946. doi:10.1016/j.enconman.2026.121946.

[44] H. Zsiborács, G. Pintér, A. Vincze, N. H. Baranyai, M. J. Mayer, The reliability of photovoltaic power generation scheduling in seventeen European countries, Energy Conversion and Management 260 (2022) 115641. doi:10.1016/j.enconman.2022.115641.

[45] J.-F. Toubeau, J. Bottieau, Y. Wang, F. Vallée, Interpretable probabilistic forecasting of imbalances in renewable-dominated electricity systems, IEEE Transactions on Sustainable Energy 13 (2022) 1267–1277. doi:10.1109/TSTE.2021.3092137.

[46] J. Han, L. Yan, Z. Li, A task-based day-ahead load forecasting model for stochastic economic dispatch, IEEE Transactions on Power Systems 36 (2021) 5294–5304. doi:10.1109/TPWRS.2021.3072904.

[47] A. Stratigakos, S. Camal, A. Michiorri, G. Kariniotakis, Prescriptive trees for integrated forecasting and optimization applied in trading of renewable energy, IEEE Transactions on Power Systems 37 (2022) 4696–4708. doi:10.1109/TPWRS.2022.3152667.

[48] D. Wahdany, C. Schmitt, J. L. Cremer, More than accuracy: end-to-end wind power forecasting that optimises the energy system, Electric Power Systems Research 221 (2023) 109384. doi:10.1016/j.epsr.2023.109384.

[49] J. M. Bates, C. W. J. Granger, The combination of forecasts, Operational Research Quarterly 20 (1969) 451–468. doi:10.2307/3008764.

[50] X. Wang, R. J. Hyndman, F. Li, Y. Kang, Forecast combinations: An over 50-year review, International Journal of Forecasting 39 (2023) 1518–1547. doi:10.1016/j.ijforecast.2022.11.005.

[51] G. Claeskens, J. R. Magnus, A. L. Vasnev, W. Wang, The forecast combination puzzle: A simple theoretical explanation, International Journal of Forecasting 32 (2016) 754–762. doi:10.1016/j.ijforecast.2015.12.005.

[52] B. Efron, The estimation of prediction error: Covariance penalties and cross-validation, Journal of the American Statistical Association 99 (2004) 619–632. doi:10.1198/016214504000000692.

[53] L. Ljung, S. Gunnarsson, Adaptation and tracking in system identification—a survey, Automatica 26 (1990) 7–21. doi:10.1016/0005-1098(90)90154-a.

[54] M. Herbster, M. K. Warmuth, Tracking the best expert, Machine Learning 32 (1998) 151–178. doi:10.1023/a:1007424614876.

[55] R. R. Bahadur, A note on quantiles in large samples, The Annals of Mathematical Statistics 37 (1966) 577–580. doi:10.1214/aoms/1177699450.

[56] A. P. Dawid, Present position and potential developments: Some personal views: Statistical theory: The prequential approach, Journal of the Royal Statistical Society. Series A (General) 147 (1984) 278–292. doi:10.2307/2981683.

[57] H. R. Künsch, The jackknife and the bootstrap for general stationary observations, The Annals of Statistics 17 (1989) 1217–1241. doi:10.1214/aos/1176347265.

[58] R. DerSimonian, N. Laird, Meta-analysis in clinical trials, Controlled Clinical Trials 7 (1986) 177–188. doi:10.1016/0197-2456(86)90046-2.

[59] S. Holm, A simple sequentially rejective multiple test procedure, Scandinavian Journal of Statistics 6 (1979) 65–70. URL: http://www.jstor.org/stable/4615733.

[60] Y. Benjamini, D. Yekutieli, The control of the false discovery rate in multiple testing under dependency, The Annals of Statistics 29 (2001) 1165–1188. doi:10.1214/aos/1013699998.

[61] Grid Controller of India Limited, Daily power supply position reports, national load despatch centre, Public file interface of grid-india.in; reports 2017–2026, accessed 14 September 2026, 2026. URL: https://grid-india.in.

[62] Central Electricity Authority, Daily generation report, national power portal, Public reports 2018–2026, accessed 14 September 2026, 2026. URL: https://npp.gov.in.

[63] Central Electricity Authority, Daily renewable generation report, renewable project monitoring division, Public reports 2019–2025, accessed 14 September 2026, 2026. URL: https://cea.nic.in.

[64] India Meteorological Department, Gridded daily rainfall (0.25°) and maximum/minimum temperature (1°) data, climate monitoring and prediction group, pune, Yearly binary files 2017–2025, accessed 14 September 2026, 2026. URL: https://imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html.

[65] R. Kumar, RE-FUSED: State-day panel of India's power system from official data, with forecasting, forecast correction and scheduling code, results and logs [dataset], Zenodo, version 1.0.0, 2026. doi:10.5281/zenodo.23301898.

[66] Grid Controller of India Limited, Deviation settlement mechanism rate files, national load despatch centre, Public file interface of grid-india.in; files 2018–2026, accessed 14 September 2026, 2026. URL: https://grid-india.in.

[67] New York Independent System Operator, Market information system public data: zonal load forecasts (isolf), integrated actual load (palintegrated), day-ahead and real-time zonal lbmp, day-ahead ancillary service prices, http://mis.nyiso.com/public/, 2026. Accessed 13 September 2026.

[68] J. Muehlenpfordt, Time series, 2020. doi:10.25832/time_series/2020-10-06, dataset.

[69] A. Trindade, ElectricityLoadDiagrams20112014, 2015. doi:10.24432/C58C86, dataset.